$TN^* - E[\sum_{t=1}^{T} \mu_t] \to$ Regret

N^* ↓ true mean (best arm) μ_t ↓ true mean (played arm)

Hoeffding $P(|\hat\mu_t - \mu| \le \epsilon) \le 2\exp\left(\frac{-2t\epsilon^2}{(b-a)^2}\right)$

Bernoulli: $a=0, b=1$ (at confidence $1-\delta$)

Successive Elimination [Best Arm Identify] $\quad \hat\mu \pm c\sqrt{\frac{\ln(2/\delta)}{2T_i(t)}}$

Max Arm Pulls $= c\sum_{i \neq i^*} \frac{1}{\Delta_i^2} \ln\left(\frac{K}{\delta}\right)$ → failure probability (choose wrong arm) ↓ hyperparam

	Regret
ε-greedy	$O(T)$
Explore-then-Commit	$O(T)$

$\Delta = \mu^* - \mu$

need $m \sim \log(1/\delta)\,\Delta^2$ samples

[m = no. of times each arm pulled in Explore] (exploration budget)

$m \sim \frac{\log T}{\Delta^2}$

(Regret: $O(T)$ for bad m) $\quad O\left(\frac{\log T}{\Delta}\right)$ only for best m

UCB, $O\left(\frac{\log T}{\Delta}\right)$ $\quad = \hat\mu + \sqrt{\frac{C\ln T}{T_i(t)}}$ (UCB) $\quad$ bonus $\ge \frac{\Delta_i}{2}$ $\quad$ bad arm pulls $\sim \frac{\log T}{\Delta^2}$

Thompson $O\left(\frac{\log T}{\Delta}\right)$ ↓ better at delayed feedback vs UCB

$\mu_i \sim Beta(\alpha,\beta)$ [ASSUMED true mean distribution]

drawn $\tilde\mu \sim f_{\alpha\beta}(\mu) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$ $\quad [\mu \in [0,1]]$

$E[\tilde\mu] = \frac{\alpha}{\alpha+\beta}$ $\quad Var[\tilde\mu] = \frac{\alpha\beta}{(\alpha+\beta)^2(1+\alpha+\beta)}$

(NOT USED IN CODE!)

$\mu_i \sim B(\alpha,\beta)$; If reward = 1 $\alpha{+}{=}1$ else $\beta{+}{=}1$

Contextual Bandits

Confidence Ellipsoid $C_t = \{\theta : (\theta-\hat\theta)^T V^{-1}(\theta-\hat\theta) \le \beta\}$ $\quad \|\theta\|_V^T \le \beta$

Exp Reward = $\langle\theta, x_a\rangle$ [regress]

sigmoid$(\langle\theta, x_a\rangle)$ [classify] → invertible

$X\theta = y$

$\hat\theta = V^{-1}b \quad V = XX^T + \lambda I = \sum_s x_s x_s^T + \lambda I \quad b = Xy = \sum_{s=1}^{n} x_s y_s$

[ridge L2 regression]

$\arg\min_\theta \left(y_s - \langle\theta, x_t\rangle\right)^2 + \lambda\|\theta\|^2$

LinUCB $\quad \hat\mu + \sqrt{\beta}\,\|x\|_{V^{-1}}$

Cauchy-Schwartz $\langle\theta, x\rangle \le \|\theta\|_V \|x\|_V$

Regret $= \sum_{t=1}^{T} \mu^* - E[\mu_a] \le cd\sqrt{T\ln T}$ $\quad (\mu^* = \langle\theta, x^*\rangle)$

CamScanner

file, db   kafka

$$V = \lambda I + \sum_s \gamma^{t-s} x x^T$$

$$b = \sum_s \gamma^{t-s} x y$$

Data Engineering

Lin TS $f(\theta \mid data) \sim N(\hat{\theta}, \alpha^2 V^{-1})$

covariance matrix

Personalization: 1 bandit/user   1 bandit /user cluster   CP (for user at item i, time t)

[doesn't scale]

Prod Realities $x^T V x + \langle \theta, x \rangle$ → scales at $d^2 K$

reward design

delayed reward

non stationary

1. filter (Approx Nearest Neighbours)
2. estimate long reward
3. at model drift down weight

Beam: both batch & stream with same code pipeline